

* Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{Where } P(B) \neq 0$$

Q. A company as total 50 employees out of this 50 employees, 35 known Python.

Q.1) What is the probability that an employees knows Python even that employees works in IT Dept.

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Formula: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{35}{50} = 0.7$

Insights: 1) 70% employees known Python.
2) Rest 30% employees doesn't know Python.

Q.2 A college as 100 student 60 student are girls 45 of that girl use laptop

Q.1) What is the probability that student use laptop given/even that student is a girl.

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Formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{45}{60} = 0.75$$

Insights:

1) 75% girls students uses laptop.